Practical-4

Problem Definition:

Designandsimulate a secure digital communication system where two data sources (D1 and D2) are transmitted over a single channel using a 2:1 Multiplexer and received using a 1:2 Demultiplexer. Before transmission, the data should be encrypted using a bitwise XOR encryption key, and at the receiver side, the same key should be used to decrypt the data.

The objective is to verify that encryption and decryption operations cancel out correctly, ensuring data integrity across transmission.

Requirements / Specifications:

- Use a 2:1 Multiplexer to select between two data sources (D1 and D2).
- Apply bitwise XOR encryption with a predefined key (K).
- At the receiver end, use a 1:2 Demultiplexer to separate the transmitted data.
- Apply the same XOR key for decryption to retrieve the original data.
- Validate through truth tables and simulation results that encryption and decryption are complementaryoperations.

Supplementary Problems:

- Modify the design for 4 datasources using a 4:1 MUX and 1:4 DEMUX.
- Add an error detection mechanism using parity bits.
- Replace XOR encryption with bit rotation or substitution encryption for enhanced security.

Tools/Technology To Be Used:

- Simulation Software: Logisim Evolution / Proteus / Multisim / Tinkercad Circuits
- Logic Components: 2:1 Multiplexer, 1:2 Demultiplexer, XOR gates
- Verification Method: Truth tables and waveform simulation

Aim:

To design1 and simulate a **secure digital communication system** using a **2:1 Multiplexer(MUX)** and **1:2 Demultiplexer(DEMUX)**, where data from two sources is transmitted over a single channel after **bitwise XOR encryption**, and then correctly recovered at the receiver using **XOR decryption with the same key**, thereby verifying that encryption and decryption operations ensure data integrity

4:1 Multiplexer and 1:4 Demultiplexer

Description:

A ***4:1 Multiplexer (MUX)*** and ***1:4 Demultiplexer (DEMUX)*** work as complementary devices in digital systems:

MUX(4:1) : Selects **oneoffour inputs (D0, D1, D2, D3)** using **select lines (S1, S0)** and sends it to a**singleoutput(Y)**.

DEMUX(1:4): Takes **oneinput (D)** and distributes it to **one of four outputs (Y0–Y3)** using the same select lines.

Boolean Expressions:

4:1 MUX Output

1:4 DEMUX Outputs $Y = S^{-}1\ S^{-}0\ D0 + S^{-}1\ S0\ D1 + S1\ S^{-}0\ D2 + S1\ S0\ D3$

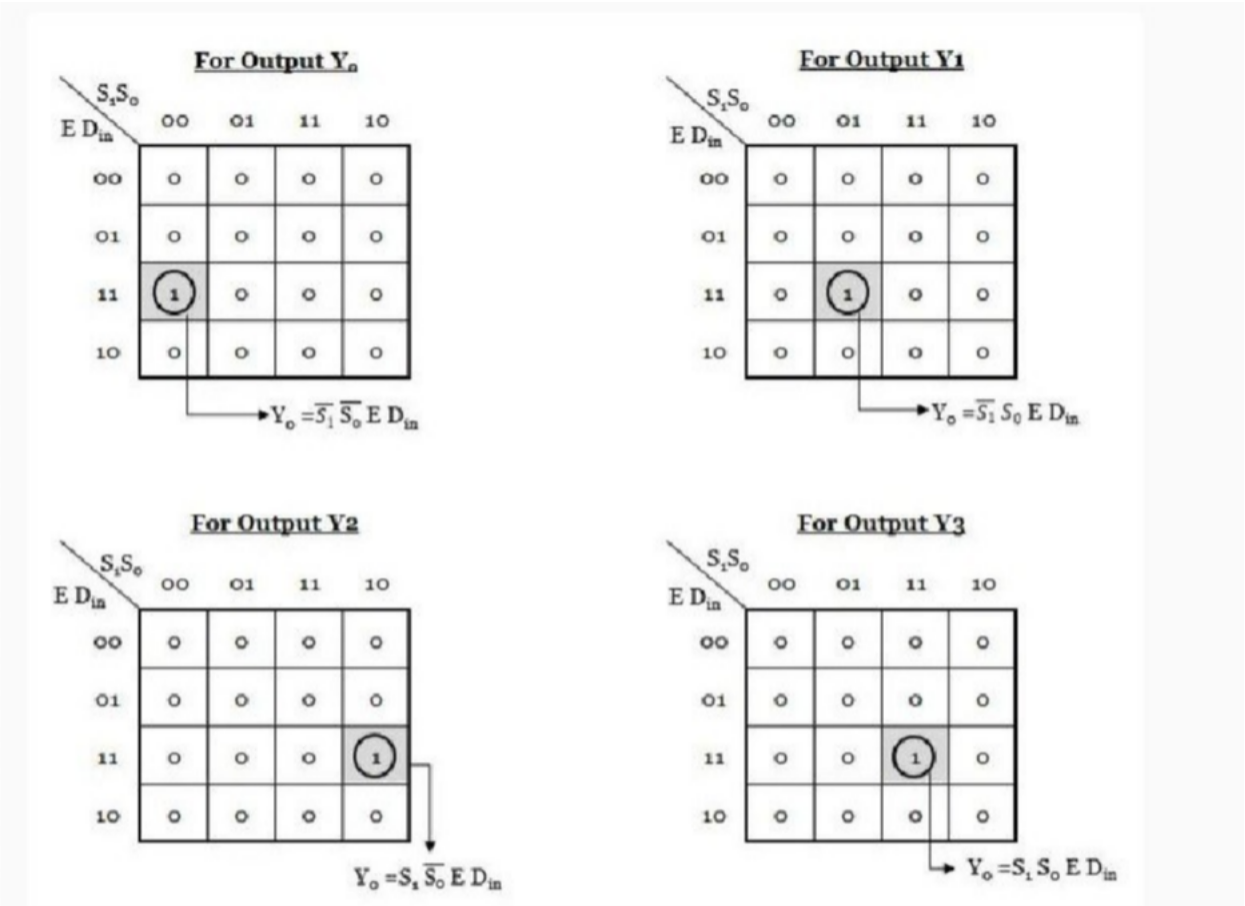
$$Y0 = D \cdot S^{-}1 \cdot S^{-}0$$
$$Y1 = D \cdot S^{-}1 \cdot S0$$
$$Y2 = D \cdot S1 \cdot S^{-}0$$
$$Y3 = D \cdot S1 \cdot S0$$

K-Map:

4:1 Multiplexer:

CD \ AB	00	01	11	10
00	0	1	1	1
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

1:4 Demultiplexer:



Truth table:

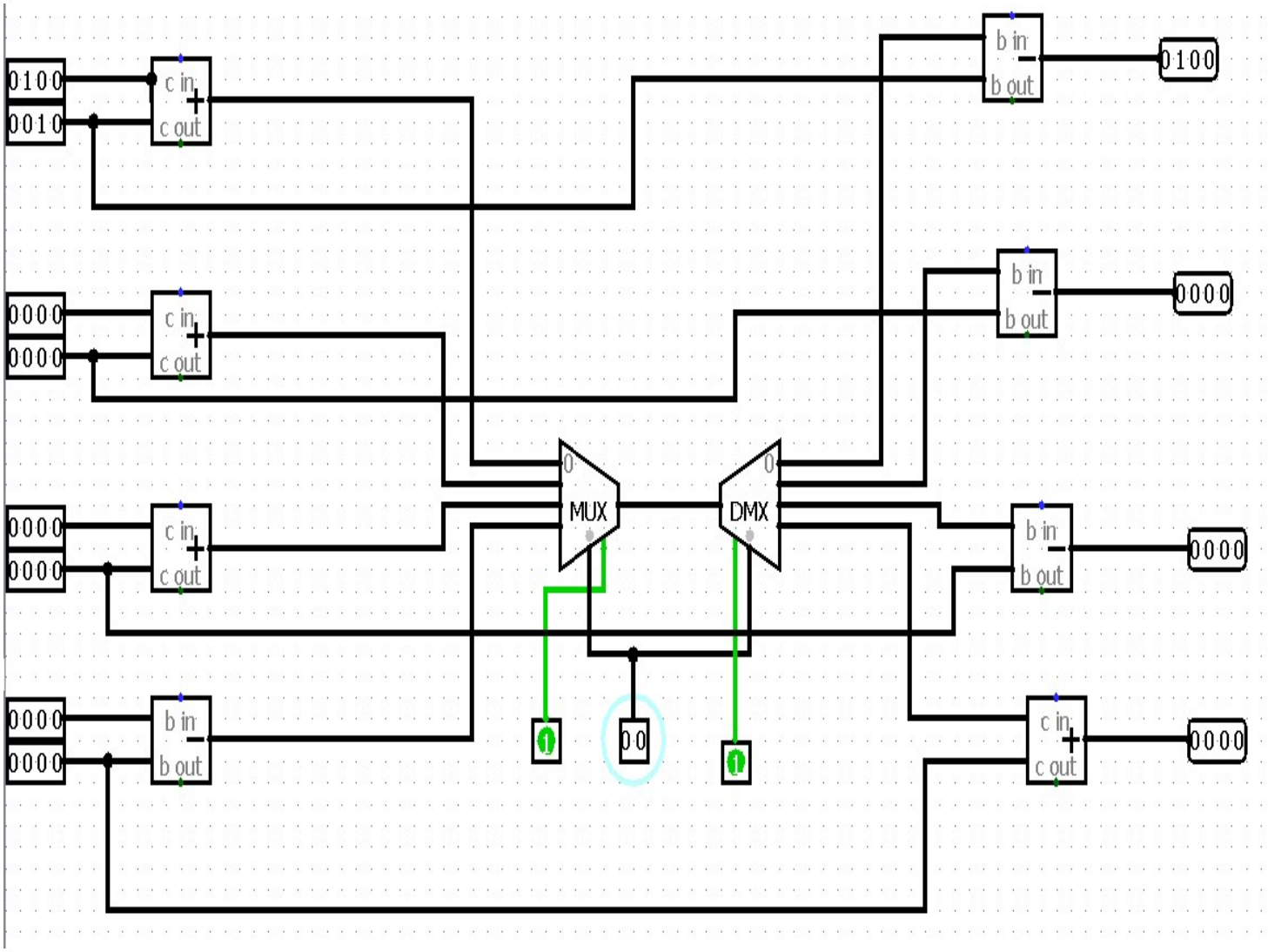
4:1 Multiplexer:

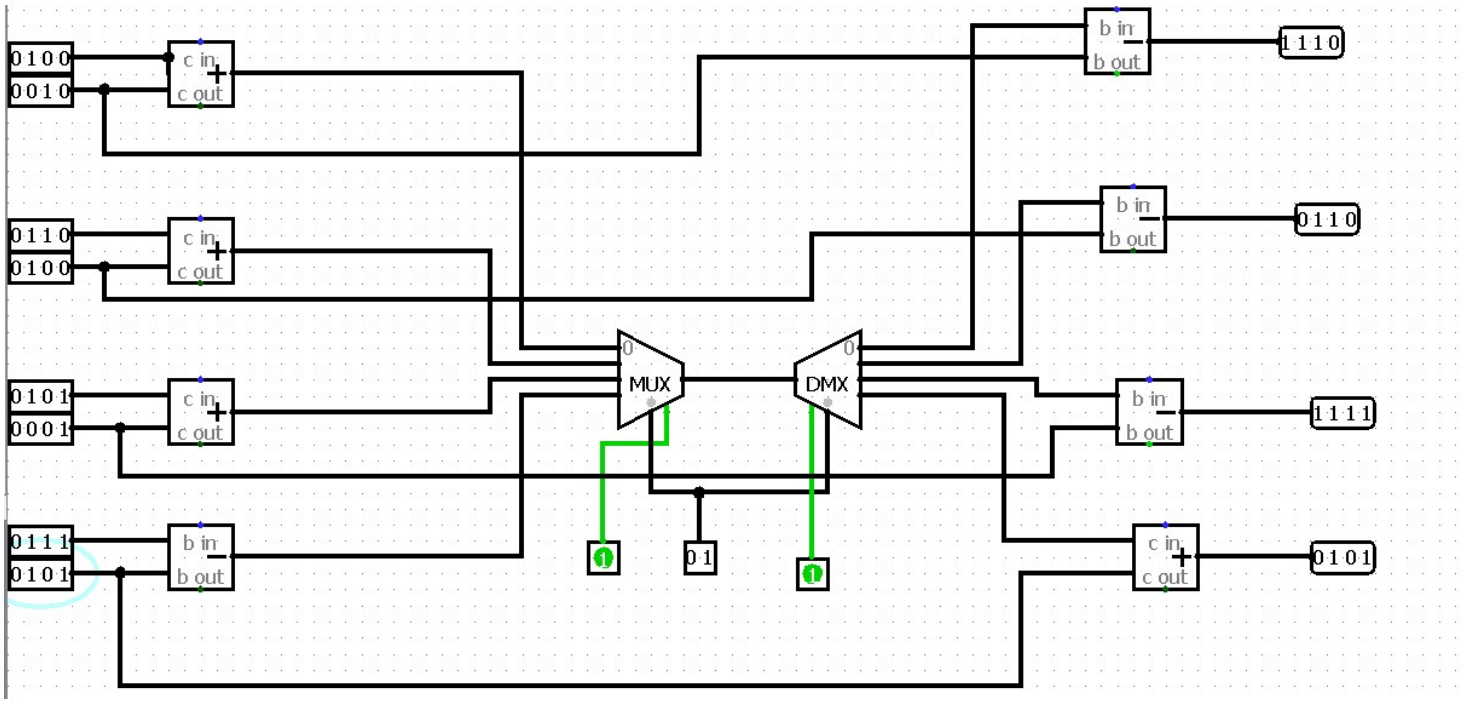
Selection Lines		Output
S0	**S1**	**Y**
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

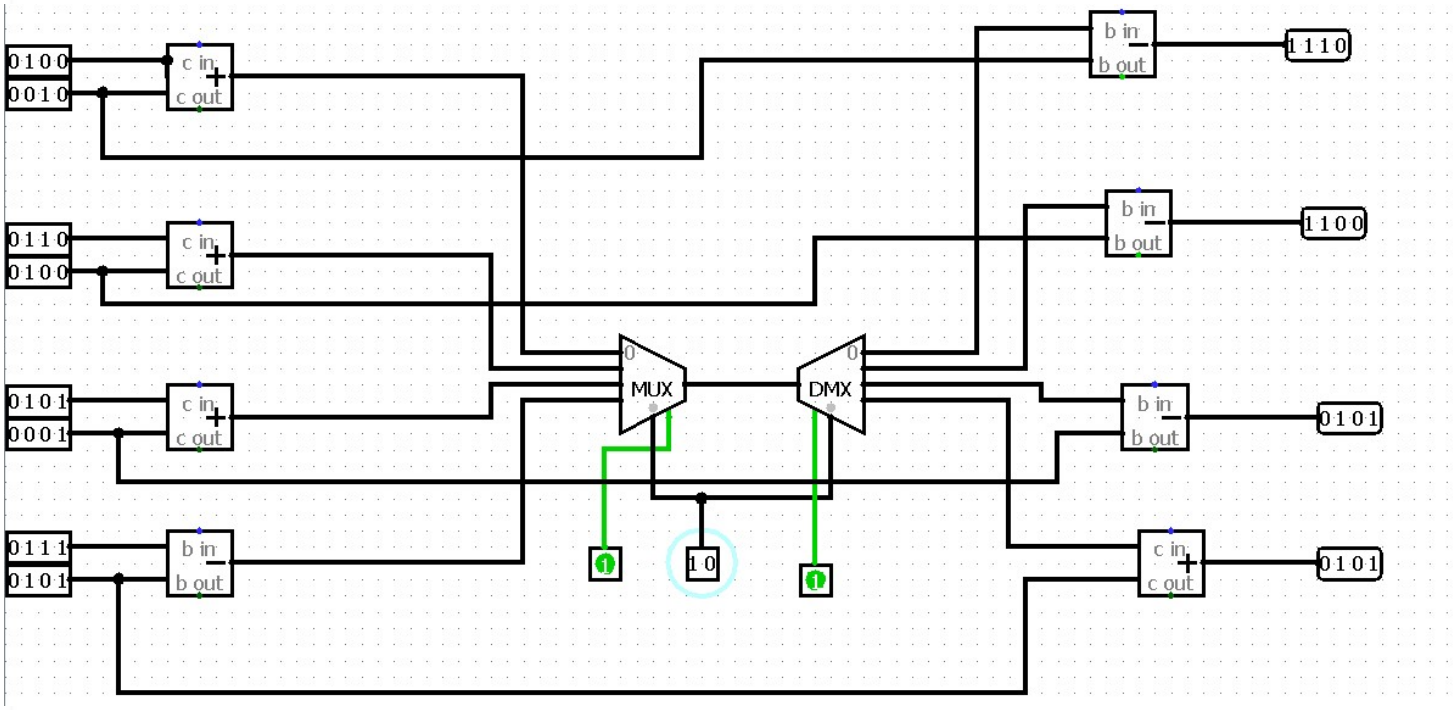
1:4 Demultiplexer:

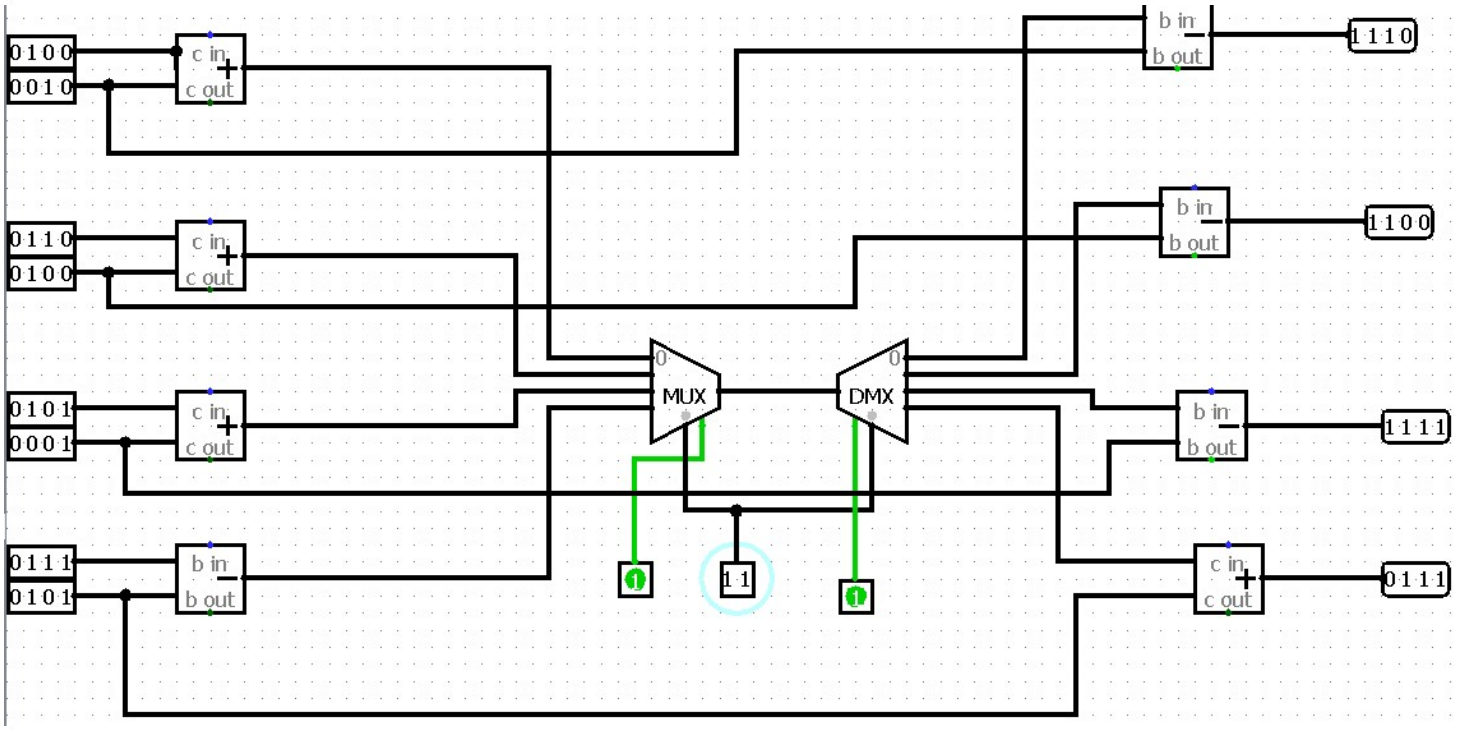
S1	S0	D	Y0	Y1	Y2	Y3
0	0	D	D	0	0	0
0	1	D	0	D	0	0
1	0	D	0	0	D	0
1	1	D	0	0	0	D

Screen shot:

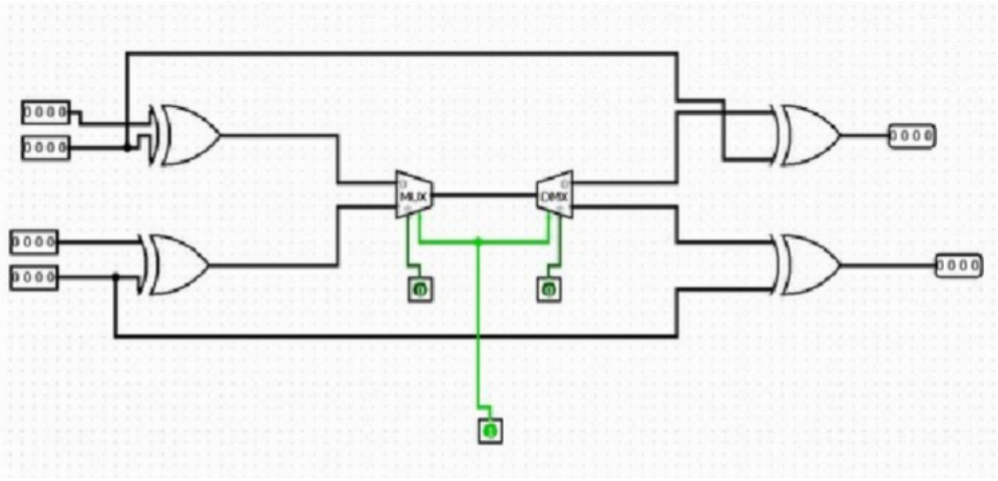

0100
0010
c in
c out
0000
0000
0000
0000
0000
0000
b in
b out
0100
0000
0000
MUX
DMX
00
c in
c out
0000

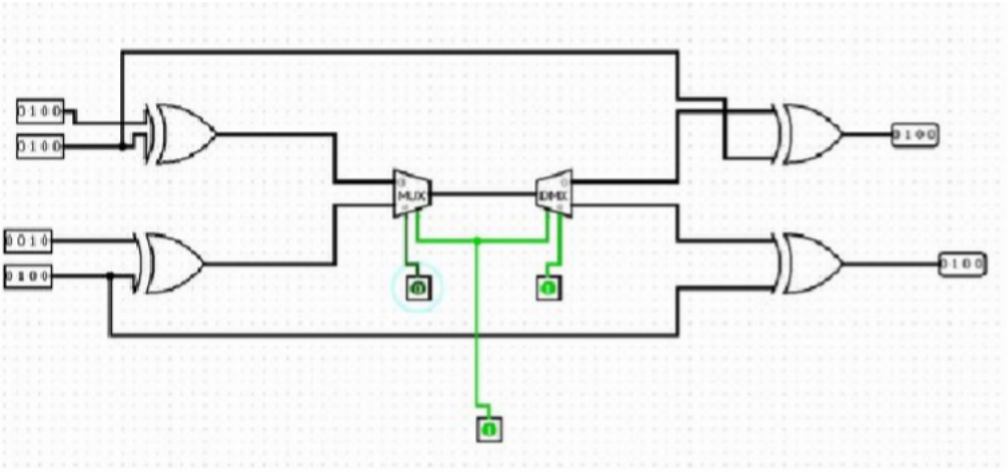

0100
0010
c in
c out
0110
0100
0101
0001
0111
0101
b in
b out
MUX
DMX
01
1110
0110
1111
0101

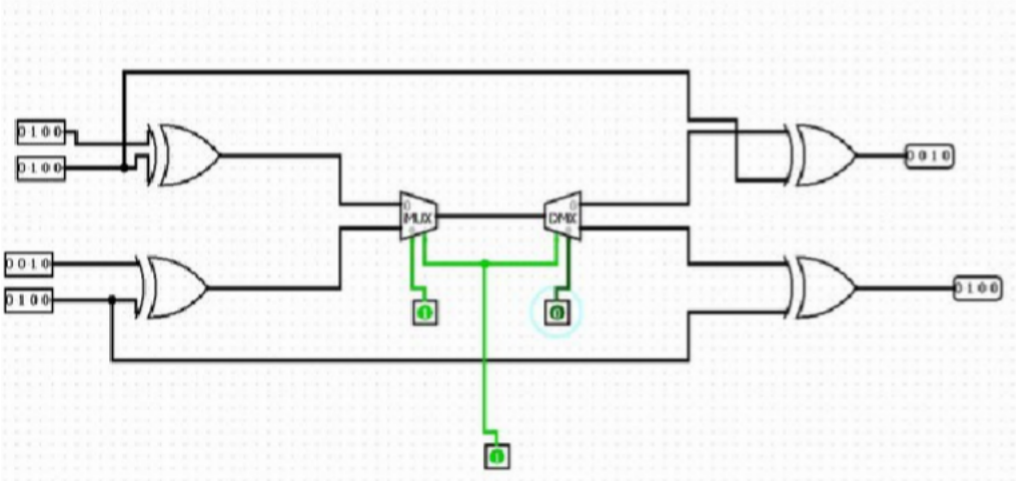

0100
0010
c in
c out
0110
0100
0101
0001
0111
0101
b in
b out
MUX
DMX
10
1110
1100
0101
0101

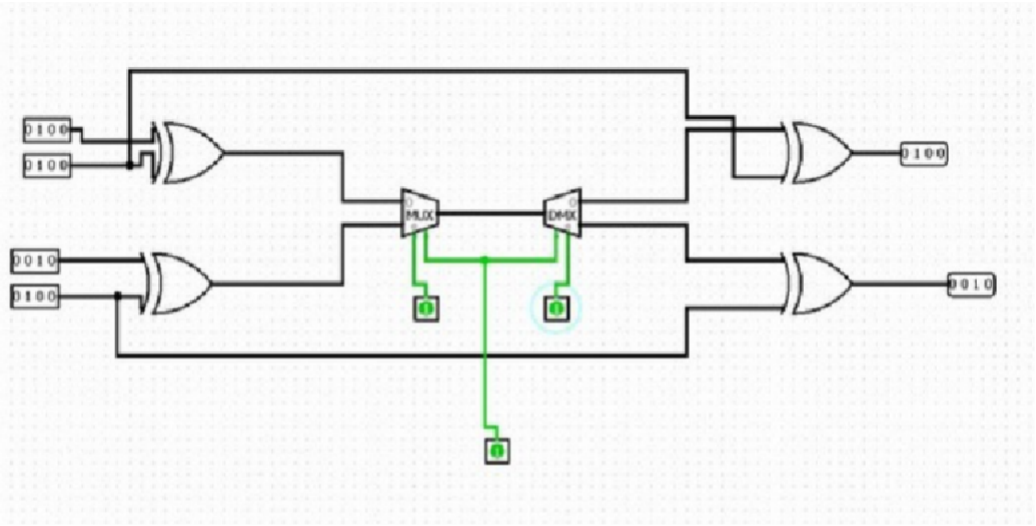

0 1 0 0
0 0 1 0
c in
c out
0 1 1 0
0 1 0 0
0 1 0 1
0 0 0 1
0 1 1 1
0 1 0 1
b in
b out
0
MUX
DMX
1 1
1 1 1 0
1 1 0 0
1 1 1 1
0 1 1 1

2:1 Multiplexer and 1:2 Demultiplexer









Conclusion: This practical session successfully demonstrates the application of multiplexing and symmetric encryption in digital communication systems. By integrating a 2:1 MUX and 1:2 DEMUX, the design proves that multiple data streams can be efficiently shared over a single physical channel without cross-talk. The implementation of the XOR gate as a stream cipher confirms the fundamental property that bitwise encryption and decryption are inverse operations when using a shared key. Ultimately, the simulation verifies that despite the encryption overhead, data integrity is maintained, highlighting how gate-level logic can be used to secure data against unauthorized access during transmission.